

NAME: _____
 TEACHER'S NAME: _____

BAULKHAM HILLS HIGH SCHOOL

TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

2005

MATHEMATICS EXTENSION 1

*Time allowed – Two hours
 (Plus five minutes reading time)*

GENERAL INSTRUCTIONS:

- Attempt ALL questions.
- Start each of the 7 questions on a new page.
- All necessary working should be shown.
- Write your teacher's name and your name on the cover sheet provided.
- At the end of the exam, staple your answers in order behind the cover sheet.
- Marks indicated for each question are only a guide and could change.

QUESTION 1

Marks

- (a) Find the acute angle between the lines $y = x$ and $y = \sqrt{3}x$. 3
- (b) If $A = (1, 4)$ and $B = (6, -12)$, find the point $P(x, y)$ which divides AB externally in the ratio $2:3$. 3
- (c) If $\cos 3x = 4 \cos^3 x - 3 \cos x$, solve $\cos 3x + 2 \cos x = 0$ for $0 \leq x \leq \pi$. 3
- (d) Find the equation of the tangent to the curve $y = \tan^2 x$ at the point $\left(\frac{\pi}{4}, 1\right)$. 3
- (e) Find the co-efficient of x^8 in the expansion $\left(\frac{2}{x} + x^3\right)^{20}$

QUESTION 2

- (a) If $f(x) = \sin x + \frac{x}{2} - 1$ has a root near $x = 0.6$, use one application of Newton's Method to find a better approximation of the root. Give your answer to 2 decimal places. 4
- (b) Evaluate $\int_0^2 \frac{x}{\sqrt{9-x^2}} dx$ using the substitution $u = 9 - x^2$. 4
- (c) Use mathematical induction to prove $5^n + 2(11)^n$ is divisible by 3 for all positive integer n such that $n \geq 1$. 3

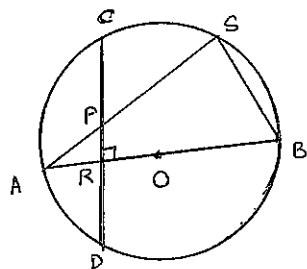
QUESTION 3

(a) Find $\int \frac{dx}{9+4x^2}$.

Marks

2

(b)



AB is a diameter and CD is perpendicular to AB.

(i) Prove PRBS is a cyclic quadrilateral.

3

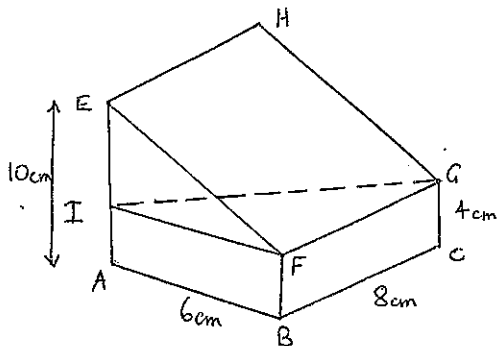
(ii) If $AP = 5$ and $AR = 4$ and $PS = 8$, find BR .

2

(c) A spherical bubble is expanding so that its volume is increasing at a constant rate of 10mm^3 per second. What is the rate of increase of its radius when its surface Area is 500mm^2 .

2

(d)

(i) Find GI .

1

(ii) Find the size of $\angle EGB$ using the Cosine Rule.

2

QUESTION 4

(a)

1

(i) Complete the table for $y = \cos^2 x$.

x	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π
y					

(ii) Sketch the curve for $y = \cos^2 x$ for $0 \leq x \leq \pi$.

1

(iii) Shaded in the region enclosed by $y = \cos^2 x$ for $0 \leq x \leq \pi$ and the line $y = 1$.

1

(iv) Find the exact area of this shaded region.

3

(v) The area below the curve $y = \cos^2 x$ and above the x axis is rotated about the x axis from $x = 0$ to $x = \pi$. Estimate this volume using the Trapezoidal Rule with 4 strips.

3

(b) (i) If $(x+1) \cdot Q(x) = x^3 + 2x^2 - 1$, find $Q(x)$.

1

(ii) Sketch the graphs of $y = x^2$ and $y = \frac{1}{x+2}$ on the same set of axes showing clearly the x coordinates for their point(s) of intersection.

3

(iii) Hence or otherwise solve $\frac{1}{x+2} > x^2$.

2

QUESTION 5

(a) A particle is moving along the x axis, its velocity V (m/s) is given by $V^2 = 16 + 4x - 2x^2$ where x is the position of the particle in metres.

(i) Show that particle is in Simple Harmonic Motion.

2

(ii) Find the amplitude and period of the motion.

3

(iii) Find the maximum speed of the particle

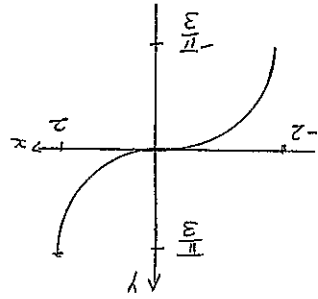
1

QUESTION 7 (Continued)

(iii) What is the probability that she wins?

2

(b)

(i) The equation of this graph is in the form $y = a \sin^{-1} bx$, find the values of "a" and "b".

2

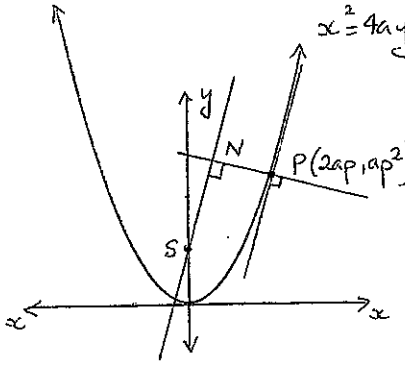
(ii) Show the gradient of the tangent at $x = 0$ is $\frac{1}{3}$.

2

(iii) For what range of values of "C" will $\frac{x}{3} = a \sin^{-1} bx - c$ have solutions?

3

QUESTION 5 (Continued)

- (b)  P is an arbitrary point on the parabola $x^2 = 4ay$ and PN is perpendicular to PS , where S is the focus.
- (i) If PN has equation $x + py = 2ap + ap^3$, find the equation of SN. 2
- (ii) Show that N has coordinates $(ap, ap^2 + a)$. 2
- (iii) Find the locus of N as P moves on the parabola. 2

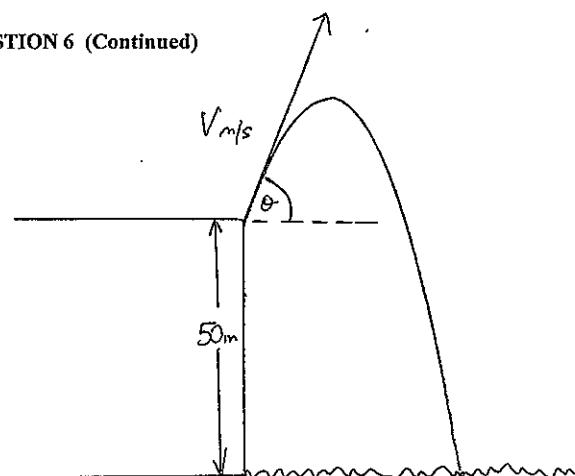
QUESTION 6

- (a) The rate at which a body's temperature (T) rises is proportional to the difference between its temperature and the surrounding medium (C).
i.e. $\frac{dT}{dt} = k(T - C)$.
- (i) Prove $T = C + A e^{kt}$ satisfies the above differential equation. 1
- (ii) If a metal bar at 25°C is placed in an oven of 300°C and its temperature rises to 100°C after 30 minutes, find the value of A and k. 2
- (iii) What will its temperature be after a further 40 minutes? 2

QUESTION 6 (Continued)

Marks

(b)



A projectile is fired from a 50 metre cliff into the sea at a velocity V metres per second with an angle of projection θ .

Let $g = 10\text{m/s}^2$

- (i) Derive the equations of motion for the horizontal and vertical components of the motion of the projectile.. 2
- (ii) If the time of flight is 5 seconds and the range of the projectile is 100 metres, find the angle of projection and the velocity of the projectile. 3
- (iii) Find the velocity of the projectile at the point of impact with the water.

QUESTION 7

- (a) Rebecca invents a game with 2 dice which have their faces coloured the same way. i.e. 3 red, 2 white and 1 black face.

She rolls both dice and wins if she throws 2 red faces and loses if she rolls 1 or more black face on the uppermost face of the dice.

If she gets neither, she rolls again and the game stops when she wins or loses.

- (i) Show that the probability of winning in 1 throw is $\frac{1}{4}$ and the probability of losing in 1 throw is $\frac{11}{36}$. 2
- (ii) What is the probability that she wins in the first or 2nd or 3rd throw. 2

1) $y = x \rightarrow m = 1$
 $y = \sqrt{3}x \rightarrow m_2 = \sqrt{3}$
 $\tan \theta = \left| \frac{\sqrt{3}-1}{1+\sqrt{3}} \right|$
 $\theta = 15^\circ$ (3)

b) (1, 4) (6, -12) Internal
 $-2:3$ (3, -12)
 $\frac{-2(6)+3(1)}{-2+3}, \frac{-2(-12)+3(4)}{-2+3}$
 $= (-9, 36)$ (2)

c) $\cos 3x + 2\cos x = 0$
 $4\cos^3 x - 3\cos x + 2\cos x = 0$
 $4\cos^3 x - \cos x = 0$
 $\cos x (4\cos^2 x - 1) = 0$
 $\therefore \cos x = 0 \quad \cos x = \pm \frac{1}{2}$
 $x = \frac{\pi}{2}, \frac{\pi}{3}, \frac{2\pi}{3}$ (3)

d) $y = \tan^2 x$
 $\frac{dy}{dx} = 2 \tan x \sec^2 x$
at $x = \frac{\pi}{4}$ $y' = 2 \tan \frac{\pi}{4} \sec^2 \frac{\pi}{4}$
 $= 2 \cdot 1 \cdot 2$
 $= 4$

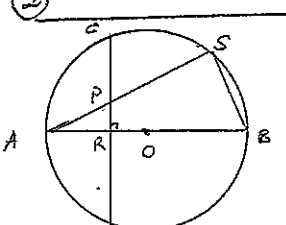
Solutions Total (92)
 $\therefore$ Tangent $y-1 = 4(x-\frac{\pi}{4})$
 $y = 4x - \pi + 1$ (3)

e) $(\frac{2}{x} + x^3)^{20}$
General term $C_k 2^{20-k} x^{20-k-20k}$
 $= C_k 2^{20-k} x^{4k-20}$
Now $4k-20 = 8$
 $4k = 28$
 $k = 7$
 $\therefore$ Coefficient is $C_7 2^{13}$ (3)

Question 2. (11)
a) $f(x) = \sin x + \frac{x}{2} - 1$
 $f(0.6) = \sin(0.6) + 0.3 - 1$
 $= -0.135 \dots$
 $f'(x) = \cos x + \frac{1}{2}$
 $f'(0.6) = 1.325 \dots$
 $\therefore a_1 = 0.6 - \frac{(-0.135 \dots)}{(1.325 \dots)}$
 $a_1 = 0.70$ (3)

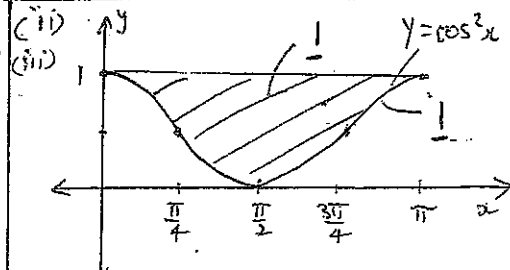
b) $\int_0^2 \frac{x}{\sqrt{9-x^2}} dx$
when $x=2 \quad u=5$
 $x=0 \quad u=9$
 $u = 9-x^2$
 $\frac{du}{dx} = -2x$
 $dx = \frac{du}{-2x}$
 $\therefore \int_9^5 \frac{x}{\sqrt{u}} \cdot \frac{du}{-2x}$
 $= -\frac{1}{2} \int_9^5 u^{-\frac{1}{2}} du$
 $= -\frac{1}{2} [2\sqrt{u}]_9^5$
 $= -\frac{1}{2} [2\sqrt{5} - 2\sqrt{9}]$
 $= -\frac{1}{2} (2\sqrt{5} - 6)$
 $= 3 - \sqrt{5}$ (4)

Prove $5^n + 2(11)^n$ is $\div$ by 3.
Step 1. Prove true for $n=1$
 $5 + 22 = 27$ which is $\div$ by 3.
Step 2. Assume true for $n=k$
 $\frac{5^k + 2(11)^k}{3} = M$ where M is an integer.
 $\therefore 5^k = 3M - 2(11)^k$
Step 3. Prove true for $n=k+1$
 $5^{k+1} + 2(11)^{k+1}$
 $= 5(5^k) + 2(11)^k \cdot 11$
 $= 5(3M - 2(11)^k) + 22(11)^k$
 $= 15M - 10(11)^k + 22(11)^k$
 $= 15M + 12(11)^k$
 $= 3(5M + 4(11)^k)$
which is $\div$ by 3.
Step 4. Prove true for $n=1$ & assumed true for $n=k$ & proven true for $n=k+1$,
 $\therefore$ true for $n=1, n=2, \dots$
& for all n by Mathematical Induction. (11)

Question 3. (12)
a) $\int \frac{dx}{9+4x^2} = \int \frac{dx}{4(\frac{9}{4} + x^2)}$
 $= \frac{1}{4} \int \frac{dx}{(\frac{3}{2})^2 + x^2}$
 $= \frac{1}{4} \left(\frac{1}{\frac{3}{2}} \right) \tan^{-1} \frac{x}{(\frac{3}{2})} + C$
 $= \frac{1}{6} \tan^{-1} \frac{2x}{3} + C$ (2)
b) 
 $\angle PRB = 90^\circ$ ($CD \perp AB$)
 $\angle ASB = 90^\circ$ (Angle in a semi-circle)
 $\therefore \angle PRB + \angle ASB = 180^\circ$
Hence $\angle SPR + \angle RBS = 180^\circ$
(Angle Sum of Quad).
Hence PRSB is a cyclic quad
(Opposite angles supplementary).

AP x AS = AR . AB
but BR = x
 $\therefore 5 \times 13 = 4 \times (x+4)$
 $65 = 4x + 16$
 $4x = 49 \Rightarrow x = \frac{49}{4}$ (4)
c) $\frac{dv}{dt} = 10$. $V = \frac{4}{3}\pi r^3$. $\frac{dV}{dr} = 4\pi r^2$
 $\frac{dv}{dt} = \frac{dV}{dr} \cdot \frac{dr}{dt}$
 $10 = 4\pi r^2 \cdot \frac{dr}{dt}$
 $\therefore 10 = 500 \cdot \frac{dr}{dt}$
 $\therefore \frac{dr}{dt} = \frac{1}{50}$ (2)
d) GI = 10 (Pythagoras)
 $EB = \sqrt{136}$ (1)
 $BG = \sqrt{80}$ (1)
 $\therefore EF = \sqrt{72}$ (1)
 $EG = \sqrt{136}$ (1)
Let $\angle EGB = x$
 $\cos x = \frac{(\sqrt{136})^2 + (\sqrt{80})^2 - (\sqrt{72})^2}{2 \cdot \sqrt{136} \cdot \sqrt{80}}$
 $x = 67.27^\circ$ (4)
4x) $y = \cos^2 x$ (15)

x	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π
y	1	$\frac{1}{2}$	0	$\frac{1}{2}$	1



(iv) Area = $\int_0^\pi 1 - \cos^2 x$
 $= \int_0^\pi \sin^2 x$
 $= \int_0^\pi \left(\frac{1}{2} - \frac{1}{2} \cos 2x\right)$
 $= \left[\frac{x}{2} - \frac{\sin 2x}{4}\right]_0^\pi$
 $= \left(\frac{\pi}{2} - \frac{\sin 2\pi}{4}\right) - (0 - 0)$
 $= \frac{\pi}{2}$

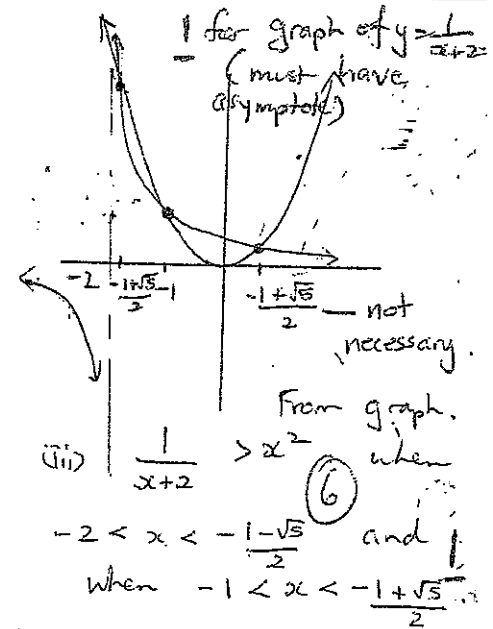
(v) $V = \pi \int y^2 dx$
 $= \pi \int_0^\pi \cos^4 x dx$

x	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
y	1	$\frac{1}{4}$	0	$\frac{1}{4}$	1

$V = \pi \left[\frac{\pi}{2} \left(1 + 1 + 2 \left(\frac{1}{4} + 0 + \frac{1}{4} \right) \right) \right]$
 $= \pi \left[\frac{\pi}{8} (3) \right]$
 $= \frac{3\pi^2}{8} (3.70)$ (10)

b) (i) $\frac{x^2 + x - 1}{x+1} = \frac{x^3 + 2x^2 + 0x - 1}{x^3 + x^2}$
 $\frac{x^2 + x - 1}{x+1} = \frac{x^2 + x}{x^2 + x} - \frac{1}{x+1}$
 $= 1 - \frac{1}{x+1}$

(ii) Pt of intersection is when $\frac{1}{x+2} = x^2$
 $x^3 + 2x^2 - 1 = 0$ from (i)
 $(x+1)(x^2 + x - 1) = 0$
 $x = -1$ or $x = \frac{-1 \pm \sqrt{5}}{2}$
 $= \frac{-1 \pm \sqrt{5}}{2}$



5b) (i) $x + py = 2ap + ap^3$ (13)
 $py = -x + 2ap + ap^3$
 $y = \frac{-x}{p} + 2a + ap^2$
 $m = -\frac{1}{p}$ $\perp m = p$
 $\therefore$ Eq'n of SN Focus (0, a)
 $y = px + a$

(ii) $x + py = 2ap + ap^3$
 $y = px + a$
 $x + p(px + a) = 2ap + ap^3$
 $x(1+p^2) = ap + ap^3$
 $x(1+p^2) = ap(1+p^2)$
 $x = ap$
 $y = ap^2 + a$
 $N(ap, ap^2 + a)$

(iv) $x = ap \Rightarrow p = \frac{x}{a}$
 $y = ap^2 + a = a \left(\frac{x}{a}\right)^2 + a$
 $y = \frac{x^2}{a} + a$
 $x^2 - ay + a^2 = 0$ (6)

(v) (i) $V^2 = 16 + 4x - 2x^2$
 $\frac{1}{2} V^2 = 8 + 2x - x^2$
 $\dot{x} = \frac{d}{dx} \left(\frac{1}{2} V^2 \right) = 2 - 2x$
 $\dot{x} = -2(x-1)$
 which is in the form $\dot{x} = -n^2(x-b)$

(i) Endpoints when $v=0$
 $0 = 16 + 4x - 2x^2$
 $x^2 - 2x - 8 = 0$
 $(x-4)(x+2) = 0$
 $x = -2, 4$
 Amplitude = 3
 Period = $\frac{2\pi}{n}$ $n=2$
 $= \frac{2\pi}{\sqrt{2}}$ or $\sqrt{2}\pi$

(iii) Max speed when $\ddot{x} = 0$
 $-2(x-1) = 0$
 $x = 1$
 $V^2 = 16 + 4 - 2 = 18$
 $V = \sqrt{18} = 3\sqrt{2}$ (7)

Question 6 (14)
 a) $\frac{dT}{dt} = k(T-c)$
 $T = C + Ae^{kt} \Rightarrow Ae^{kt} = T - C$
 $\frac{dT}{dt} = Ake^{kt} = k(T-c)$

$\therefore T = C + Ae^{kt}$ is a solution to $\frac{dT}{dt} = k(T-c)$
 (ii) when $t=0$ $T=25$ $C=300$
 $25 = 300 + Ae^0$
 $A = -275$
 $T = 300 - 275e^{kt}$
 when $t=30$ $T=100$
 $100 = 300 - 275e^{30k}$
 $275e^{30k} = 200$
 $30k = \ln\left(\frac{200}{275}\right)$
 $k = \frac{\ln\left(\frac{200}{275}\right)}{30} = -0.0106$

(iii) Find T when $t = 70$
 $T = 300 - 275e^{70(-0.0106)}$
 $T = 169.2$ (6)

6b) (i) $\ddot{x} = 0$
 $\ddot{x} = c_1$
 $\dot{x} = V \cos \theta$
 $x = Vt \cos \theta + c_2$
 when $t=0$ $x=0 \therefore c_2 = 0$
 $\therefore x = Vt \cos \theta$

6b) $y = -10$
 $y = -10t + c_3 = V \sin \theta$
 $y = -10t + V \sin \theta$
 $y = -5t^2 + V \sin \theta t + c_4$
 when $t=0, y=50 \Rightarrow c_4 = 50$

(ii) $y = -5t^2 + V \sin \theta t + 50$
 when $t=5, x=100$ and $y=0$

$\therefore 100 = 5V \cos \theta$
 $V \cos \theta = 20 \quad \text{--- (1)}$

$0 = -125 + 5V \sin \theta + 50$

$75 = 5V \sin \theta$

$15 = V \sin \theta \quad \text{--- (2)}$

② $\div$ ① $\frac{3}{4} = \tan \theta$

$\theta = 36.52^\circ$

$\therefore V \cos 36.52^\circ = 20$

$V = \frac{20}{\cos 36.52^\circ}$

$= 25 \text{ m/s}$

(iii) At impact $V = \sqrt{(u_x)^2 + (u_y)^2}$

when $t=5, V=25$

$x = 25 \cos 36.52^\circ$
 $= 20 \text{ m/s}$

$y = -10(5) + 25 \sin 36.52^\circ$
 $= 25 \text{ m/s}$

$V = \sqrt{(20)^2 + (25)^2}$
 $= \sqrt{25} \text{ m/s} = 5 \text{ m/s}$

(i) $P(\text{Winning}) = \frac{3}{6} \times \frac{3}{6} = \frac{1}{4}$

$P(\text{Losing}) = P(B) + P(\text{other Black})$
 $= \frac{1}{6} + \frac{5}{6} \times \frac{1}{6}$
 $= \frac{11}{36}$

(ii) $P(\text{Win first go}) = \frac{1}{4}$

$P(\text{Win 2nd}) = P(\text{Draw then win})$

$= \left(\frac{4}{9}\right) \times \frac{1}{4}$

$P(\text{Win 3rd}) = P(\text{Draw} \rightarrow \text{Draw} \rightarrow \text{Win})$

$= \frac{4}{9} \times \frac{4}{9} \times \frac{1}{4}$

$= \left(\frac{4}{9}\right)^2 \times \frac{1}{4}$

$\therefore P(\text{Win 1st, 2nd, 3rd...}) =$

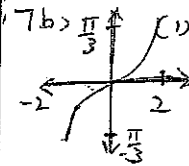
$\frac{1}{4} + \frac{1}{4} \times \frac{4}{9} + \frac{1}{4} \times \left(\frac{4}{9}\right)^2 = 1$

(iii) $P(\text{Wins}) =$

$\frac{1}{4} \left(1 + \frac{4}{9} + \left(\frac{4}{9}\right)^2 + \left(\frac{4}{9}\right)^3 + \dots\right)$

$= \frac{1}{4} \left(\frac{1}{1 - \frac{4}{9}}\right)$

$= \frac{1}{4} \times \frac{9}{5} = \frac{9}{20}$



$y = a \sin^{-1} bx$

range $-\frac{\pi}{3} \leq \frac{\pi}{3}$

Now $y = \sin^{-1} x$

range $-\frac{\pi}{2} \leq \frac{\pi}{2}$

$\frac{\pi}{3} \leq \frac{2}{3} \leq \frac{\pi}{2} \therefore a = \frac{2}{3}$

$y = \sin^{-1} bx$

$x = \frac{1}{b} \sin \frac{y}{a}$

range is $-2 \leq 2$ (twice that of $y = \sin^{-1} x$)

$\therefore \frac{1}{b} = 2$

$b = \frac{1}{2}$

(iv) $y = \frac{2}{3} \sin^{-1} \frac{x}{2}$

$y' = \frac{2}{3} \cdot \frac{1}{\sqrt{1 - \left(\frac{x}{2}\right)^2}}$

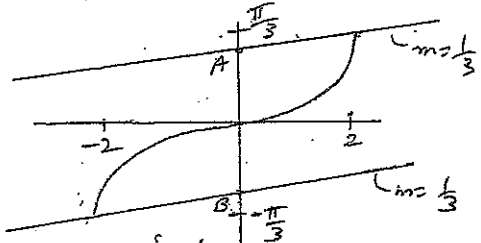
$= \frac{2}{3} \cdot \frac{1}{\sqrt{4 - x^2}}$

$= \frac{2}{3} \cdot \frac{1}{\sqrt{4 - x^2}}$

$= \frac{2}{3\sqrt{4 - x^2}}$

when $x=0, y' = \frac{1}{3}$

(ii) $\frac{x}{3} + c = \frac{2}{3} \sin^{-1} \frac{x}{2}$



$\therefore$ Require to solve $y = \frac{x}{3} + c$ & $y = \frac{2}{3} \sin^{-1} \frac{x}{2}$ simultaneously

C will lie between A and B.

At A $y = \frac{x}{3} + A$ passes through

$(2, \frac{\pi}{3})$

$\therefore \frac{\pi}{3} = \frac{2}{3} + A$

$\therefore A = \frac{\pi}{3} - \frac{2}{3}$

At B $y = \frac{x}{3} + B$ passes through

$(-2, -\frac{\pi}{3})$

$-\frac{\pi}{3} = \frac{-2}{3} + B$

$\therefore B = \frac{2 - \pi}{3}$

$\therefore \frac{2 - \pi}{3} \leq c \leq \frac{\pi - 2}{3}$

1 mark for recognising need to solve $y = \frac{x}{3} + c$ & $y = \frac{2}{3} \sin^{-1} \frac{x}{2}$ simultaneously

6